

Lesson plan (template – 2)

Name of the Programme:	B.Sc(H)	
Title of the Course:	Python Programming for Data Analytics	
Course code	CS1280	
Semester:	II	
Names of the Course Instructor(s)	Prof. Muralidhar Billa Prof. Ayesha Mafrah A M	
Course credit	No. of hours per week (L + T + P) (1:0:4)	Total no. of Teaching hours
3	5	75

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/ reference
1	1	Introduction to Python: history, features, environment setup	Lecture: Overview of Python, uses, installing Python/Jupyter.	Zelle Ch.1
2,3	1	Basic Syntax & Variables	Practical: Write first script, variables, expressions.	Zelle Ch.2
4,5	1	Basic Programming (Lab Program 2A–C)	Practical: Simple calculator, temperature converter, largest among 3 numbers.	Lab Manual 2
6	1	Control Structures: if, elif, else	Lecture: Conditional logic with examples.	Zelle Ch.3
7,8	1	Conditional Programs	Practical: Decision-making programs.	Lab Manual
9,10	1	Loops: for, while, nested loops	Practical: Pattern programs, counters.	Zelle Ch.4

11	1	Functions & Modules	Lecture: Defining functions, parameters, return values.	Zelle Ch.5
12,13	1	Functions & Recursion (Lab Program 3A–B)	Practical: Factorial recursion, word count.	Lab Manual 3
14,15	1	Integrated Problem Solving	Practical: Combine loops, conditions, functions.	Instructor Notes
16	2	Data Structures Overview	Lecture: Lists, tuples, sets, dictionaries.	Zelle Ch.7
17,18	2	Lists (Lab Program 4A)	Practical: Insert, delete, search operations.	Lab Manual 4
19,20	2	Dictionaries (Lab Program 4B)	Practical: Merge dictionaries, comprehension.	Lab Manual
21	2	Arrays & 2D Arrays	Lecture: 1D/2D arrays, nested loops.	Zelle Ch.8
22,23	2	Array Operations	Practical: 2D array exercises, grid problems.	Instructor Notes
24,25	2	Classes & Objects	Practical: Create classes, objects, methods.	Zelle Ch.10
26	2	File Handling Concepts	Lecture: Open/read/write files, CSV structure.	McKinney Ch.3

27,28	2	CSV Handling (Lab Program 5)	Practical: Read CSV, analyze contents.	Lab Manual 5
29,30	2	Exception Handling	Practical: try/except programs, custom exceptions.	Zelle Ch.11
31	3	Introduction to NumPy	Lecture: Arrays, indexing, slicing, shapes.	McKinney Ch.4
32,33	3	NumPy Arrays (Lab Program 6A)	Practical: Create/reshape arrays.	Lab Manual 6
34,35	3	NumPy Matrix Operations (Lab Program 6B)	Practical: Matrix addition, multiplication.	Lab Manual
36	3	Introduction to pandas	Lecture: Series, DataFrames, indexing.	McKinney Ch.5
37,38	3	pandas DataFrame (Lab Program 7A)	Practical: Create DataFrame from CSV/dict.	Lab Manual 7
39,40	3	Data Cleaning (Lab Program 7B)	Practical: Handle missing values, duplicates.	Lab Manual
41	3	Descriptive Statistics	Lecture: Mean, median, variance, SD, distribution.	McKinney Ch.7
42,43	3	Descriptive Stats (Lab Program 8)	Practical: describe(), quantile(), var(), std().	Lab Manual
44,45	3	Data Distribution (Lab Program 9)	Practical: skew, kurtosis, value_counts.	Lab Manual

46	4	Data Cleaning & Preprocessing	Lecture: Missing data, duplicates, IQR, outliers.	VanderPlas Ch.3
47,48	4	Missing Data Handling	Practical: dropna(), fillna().	Lab Manual
49,50	4	Categorical Encoding (Program 10)	Practical: LabelEncoder, OneHotEncoder.	Lab Manual 10
51	4	Scaling & Normalization	Lecture: Min-max, z-score, normalization.	McKinney Ch.7
52,53	4	Scaling (Program 11)	Practical: StandardScaler, Normalizer.	Lab Manual 11
54	4	Outlier Theory	Lecture: IQR, visualization techniques.	Instructor Notes
55,56	4	Outlier Detection (Program 12)	Practical: Boxplot, IQR detection.	Lab Manual 12
57	4	Advanced Pandas Operations	Lecture: merge, join, concat, groupBy.	McKinney Ch.8
58,59	4	Advanced Pandas (Program 13)	Practical: merging, grouping, case study.	Lab Manual 13
60	5	Introduction to Data Visualization	Lecture: Matplotlib basics, charts, customization.	McKinney Ch.9
61,62	5	Matplotlib (Lab Program 12A)	Practical: line, bar, scatter, histogram.	Lab Manual

63,64	5	Seaborn Basics	Practical: boxplots, violin plots, pairplots.	VanderPlas Ch.4
65	5	Advanced Seaborn Visualization	Lecture: Themes, palettes, heatmaps.	VanderPlas Ch.4
66,67	5	Correlation Heatmaps (Program 15)	Practical: Heatmap visualization for correlations.	Lab Manual
68,69	5	Plotly Introduction	Practical: Interactive plots.	Plotly Docs
70,71	5	Identifying Trends	Practical: Line charts, moving averages.	Lab Manual
72,73	5	Relationship Visualization	Practical: Scatter plots & regression lines.	Lab Manual
74,75	5	Mini Project (EDA Case Study)	Practical: Real-world dataset analysis (finance/health/sports).	Instructor Notes

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1	
- IBM SkillsBuild Courses: Graded Component-1	10 marks out of 70
- Quiz Questions Graded Component 2	10 marks out of 70
CIE-2	25 marks out of 70
- Theory	
CIE-3	
- Lab record writeup & viva	5 marks out of 70
- Lab Project & Viva	20 marks out of 70

Semester End Exams (SEE)

Mode of Exam	Practical
Weightage of exam	30

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1: Apply Python programming fundamentals and data structures to build basic programs, handle files, and organize data.	3	2	2	3	1	-	1	2	-	-	-	-
CO2: Implement data analysis using NumPy and Pandas, including descriptive statistics and data preprocessing techniques.	3	2	1	3	1	-	1	2	-	1	-	-
CO3: Create data visualizations using Matplotlib and Seaborn to represent data insights effectively.	3	3	2	3	1	1	2	2	-	-	-	-
CO4: Develop and present a complete data analysis project demonstrating skills in programming, data manipulation, and visualization.	3	3	2	3	1	-	1	2	-	-	-	-

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	IBM SkillsBuild Courses:	Certificate of completion for 10 marks. The specified courses on the IBM SkillsBuild platform to be uploaded to

	Graded Component-1 (online certification)	the designated Google Classroom. (Refer Table-1)
	Quiz Questions Graded Component 2	Quiz questions will be conducted for 10 marks. Total 10 Questions will be given each right answer carries one mark. No negative marking for wrong answers.

Table-1:

Graded Component-1 (10 Marks):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	Excellent (10 Marks)	Satisfactory (5 Marks)	Needs Improvement (0 Marks)
Completion of Courses (10 Marks)	Successfully completes all specified courses on the IBM SkillsBuild platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Graded Component-2 (10 Marks):

Criteria	Correct answer (1 Marks)	Wrong Answer (0 Marks)
Quiz questions (10 Marks)	Successfully choosing the correct option Student will be awarded (1 marks).	Fails to choose correct option will be reward as (0 marks).

Note: Total 20 Questions will be given Each right answer carries 1 mark. No negative marking for wrong answers

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks	25
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CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab record+ Viva = 5 marks (<i>Refer Table-2</i>) Lab Project and Viva = 20 marks (<i>Refer Table-3</i>)

Total Marks		30
SEE Component		Rubrics for Assessment
SEE	Practical Test	Lab Practical Test = 15 marks (<i>Refer Table-4</i>)

Note: Overall lab tests will be conducted, including 2 programs where Write-up: 10 Marks & Viva-Voce: 5 Marks will sum up to 30 M.

Table-2

Lab Program Submission (Google Classroom): 6 marks

Sl. No	Criteria	Measuring Method	Excellent	Very Good	Good	Average	Needs Improvement
1	Code Quality & Logic(3)	Review of code in Google Classroom submission	Code is efficient, well-structured, uses correct logic; no errors(3)	Minor issues but overall correct and efficient(2)	Logic partially correct; minor errors (1)	Code runs with multiple errors; logic unclear (0.5)	Major issues; code does not run (0)
2	Output Accuracy(2)	Program execution and screenshots	Accurate output for all test cases (2)	Minor deviations in output (2)	Correct for basic cases only (1)	Output partially correct (0.5)	Output incorrect (0)
3	Timely Submission(1)	Timestamp in Google Classroom	Submitted before deadline (1)	Submitted on deadline (0.8)	Submitted within 12 hours after deadline (0.6)	Submitted within 24 hours after deadline (0.4)	Not submitted (0)

Viva: 4 Marks One mark questions will be conducted for 4 marks. Each right answer carries one mark. No negative marking for wrong answers.

Note: The overall lab record, including 16 programs where the write-up is 6 marks & viva-voce is 4 marks, will sum up to 160 marks and will be reduced to 5 marks.

Table-3

Project rubrics (Max:20 marks)				
Sl. No	Criteria	Excellent	Good	Poor
1	Implementation (5 Marks)	Identified a dataset, problem statement and effectively communicates the expected analysis 4-5 Marks	Identified a dataset and problem statement, communicates the expected analysis with minor areas of improvement 3-1 Marks	No review is provided 0 Marks
2	Report (5 Marks)	Report is well-written, organized, and thoroughly covers all necessary aspects with clarity. 4-5 Marks	Report is mostly clear and well-structured, with minor areas for improvement. 3-1 Marks	No report provided. 0 Marks
3	Presentation (5 Marks)	Engaging, clear, and well-structured presentation that effectively communicates key points. 4-5 Marks	Presentation is mostly clear and organized, with minor areas for improvement. 3-1 Marks	Presentation is missing or significantly incomplete. 0 Marks
4	Viva (5 Marks)	Demonstrates a profound understanding of the subject matter, answering questions with clarity and depth. 4-5 Marks	Answers questions accurately and confidently, with minor gaps in understanding. 3-1 Marks	Unable to assess subject knowledge due to incomplete or missing VIVA. 0 Marks

Table-4

SEE Test rubrics (Max: 30 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor

1	Write-up (5 Marks)	Observations	Well-organized, clear, with proper logic of the problem. Validations with all test cases are handled. (5 M)	Well-organized, clear, with partial logic of the problem and validations. (1 - 4 M)	The student has not executed the program. (0 M)
2	Execution and Format of Results (5 Marks)	Observations	Students demonstrate the execution of the program with error correction and show performance efficiency. Appropriate validations Execution Results should be show in proper format (5 M)	Students demonstrate the execution of the program without error Correction and appropriate validations with a test case (minimum 1) are handled. Execution Results in without format (1- 4 M)	Execution Results with no comments and no output (0 M)
Viva Voce Rubrics (Max: 5 marks)					
1	Conceptual Understanding & Communication of Concepts (5 Marks)	Viva Voce	Explains thoroughly the program executed along with related concepts. (5 M)	Adequately explains the program executed along with related concepts (3 – 4 M)	Unable to explain the program executed along with related concepts (1-2 M)

Note: Overall lab tests will be conducted, including 2 programs where Write-up: 10 Marks & Viva-Voce: 5 Marks will sum up to 30 M.

Signature of the Course Faculty